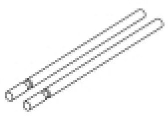
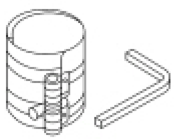
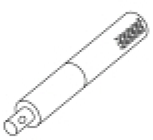
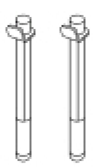
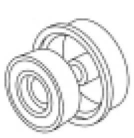
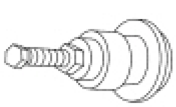
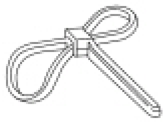
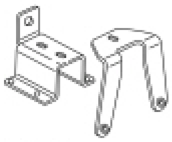
Engine

Ford Ecat Electronic Parts Catalog

Special Tool(s)

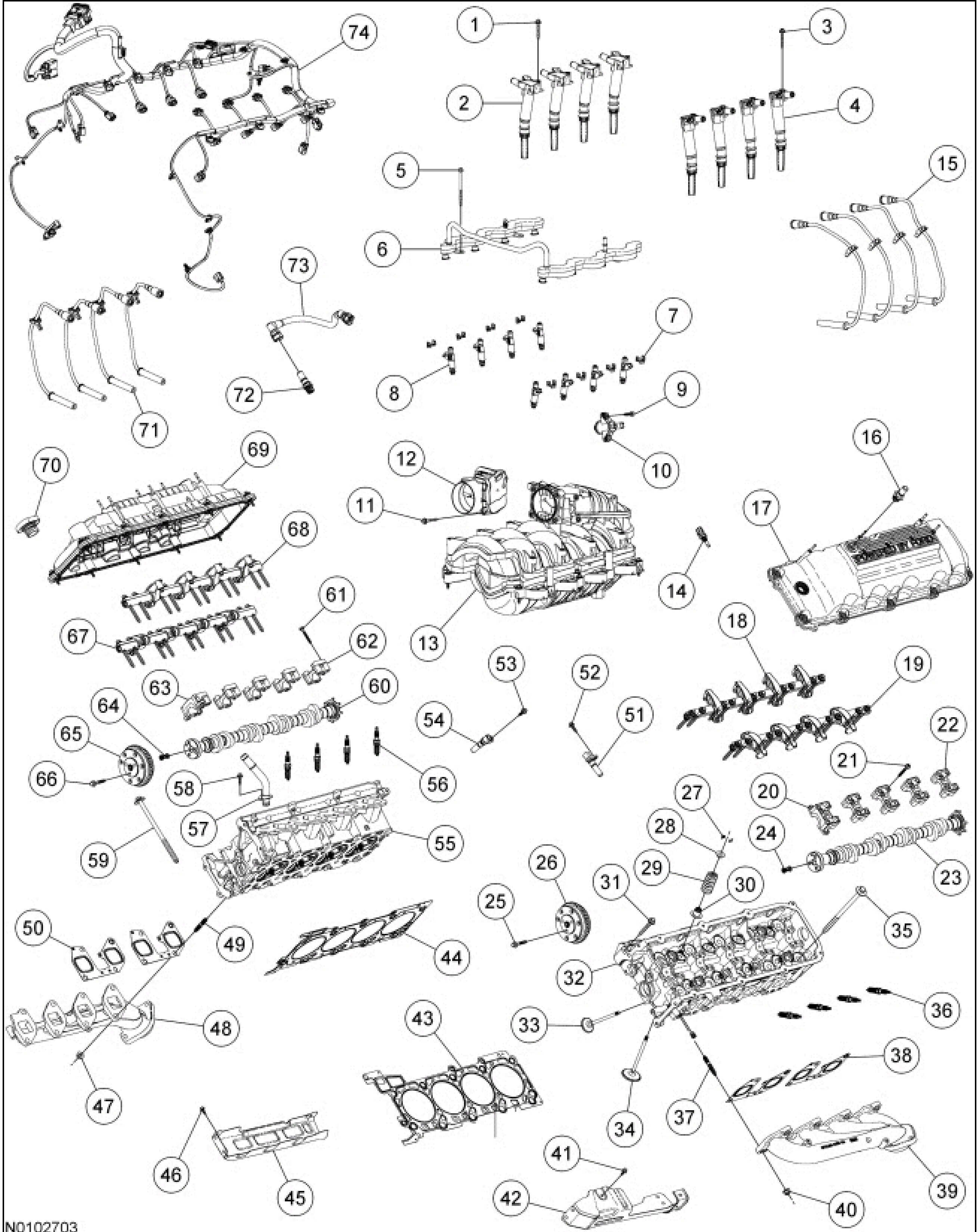
 ST2806-A	Alignment Pins, Cylinder Head 303-1040 (SR-015486)
 ST1376-A	Compressor, Piston Ring 303-D032 (D81L-6002-C) or equivalent
 ST1326-A	Handle 205-153 (T80T-4000-W)
 ST1337-A	Installer, Connecting Rod 303-442 (T93P-6136-A)
 ST3245-A	Installer, Crankshaft Damper and Crankshaft Front Seal 303-1508
 ST2980-A	Installer, Rear Main Seal 303-1250
 ST1328-A	Installer, Front Cover Oil Seal 303-335 (T88T-6701-A)

 ST1377-A	Lifting Bracket, Engine 303-F047 (014-00073) or equivalent
 ST1436-A	Strap Wrench 303-D055 (D85L-6000-A) or equivalent
 ST3110-A	Support Brackets, Engine 303-1507 or equivalent

Material

Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A
Motorcraft® Silicone Gasket Remover ZC-30-A	—
Motorcraft® Orange Concentrated Antifreeze/Coolant VC-3-B (US); CVC-3-B2 (Canada)	WSS-M97B44-D
Motorcraft® Silicone Gasket and Sealant TA-30	WSE-M4G323-A4

Engine — Upper End

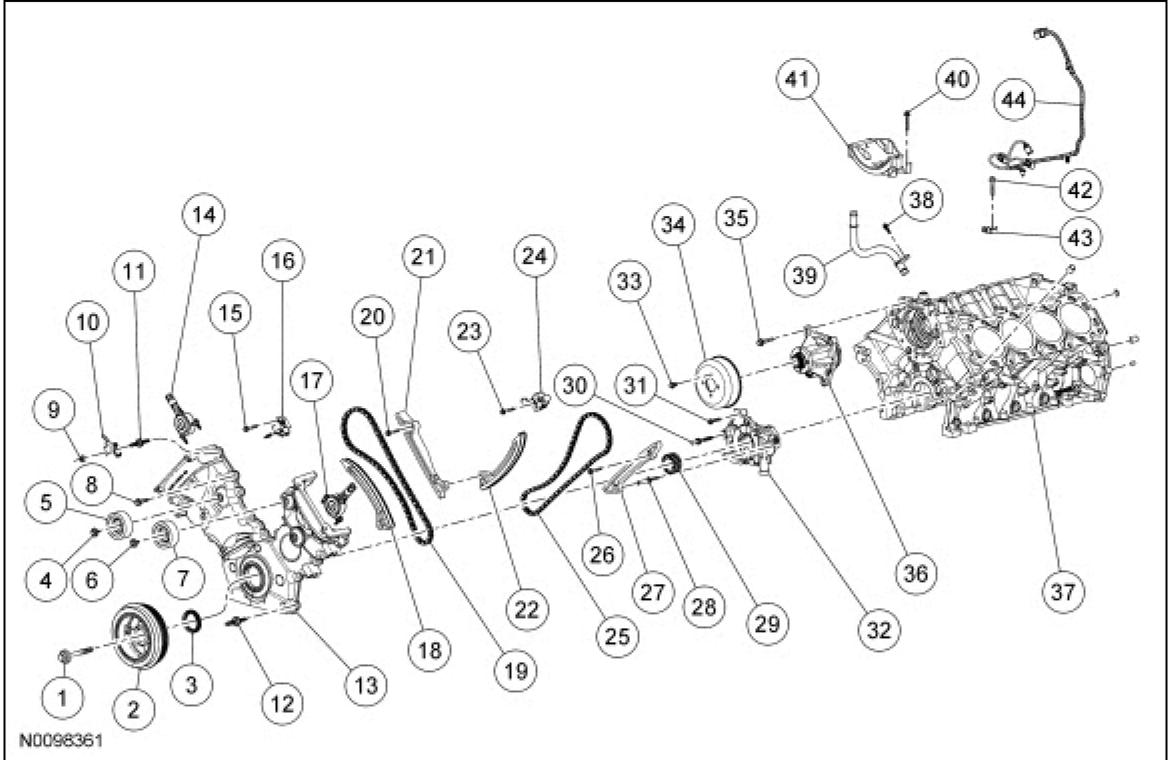


Item	Part Number	Description
1	W503282W503282	Ignition coil bolt — RH (4 required)
2	1202912029	Ignition coil — RH (4 required)
3	W503282W503282	Ignition coil bolt — LH (4 required)
4	1202912029	Ignition coil — LH (4 required)
5	W714633W714633	Fuel rail bolt (4 required)
6	9F7929F792	Fuel rail
7	9C9959C995	Fuel injector clip (8 required)

8	9F5939F593	Fuel injector (8 required)
9	W503278W503278	EVAP purge valve bolt (2 required)
10	9E9269E926	EVAP purge valve
11	W709168W709168	Throttle Body (TB) bolt (4 required)
12	9F9919F991	TB
13	94249424	Intake manifold assembly
14	6G0046G004	Cylinder Head Temperature (CHT) sensor
15	1228612286	Ignition wire — LH (4 required)
16	67626762	Crankcase ventilation tube fitting
17	6A5056A505	Valve cover — LH
18	6C5536C553	Intake rocker arm and shaft assembly — LH
19	6C5526C552	Exhaust rocker arm and shaft assembly — LH
20	6B2806B280	Front camshaft bearing cap — LH
21	W714203W714203	Camshaft bearing cap bolt — (10 required)
22	6B2846B284	Camshaft bearing cap — LH (4 required)
23	6C2556C255	Camshaft — LH
24	60526052	Variable Camshaft Timing (VCT) system oil filter
25	W712998W712998	Camshaft phaser and sprocket bolt — LH
26	6C5246C524	Camshaft phaser and sprocket — LH
27	65186518	Valve spring retainer key (32 required)
28	65146514	Valve spring retainer (16 required)
29	65136513	Valve spring (16 required)
30	6A5176A517	Valve stem seal (16 required)
31	W705738W705738	Cylinder head bolt — LH
32	60506050	Cylinder head — LH
33	65056505	Exhaust valve (8 required)
34	65076507	Intake valve (8 required)
35	60656065	Cylinder head bolt — LH (10 required)
36	1240512405	Lower spark plug (8 required)
37	W714869W714869	Exhaust manifold stud — LH (8 required)
38	94489448	Exhaust manifold gasket — LH (2 required)
39	94319431	Exhaust manifold — LH
40	W714870W714870	Exhaust manifold nut — LH (8 required)
41	W714133W714133	Exhaust manifold heat shield bolt (2 required)
42	9A4629A462	Exhaust manifold heat shield
43	60836083	Cylinder head gasket — LH
44	60516051	Cylinder head gasket — RH
45	9A4629A462	Exhaust manifold heat shield
46	W714133W714133	Exhaust manifold heat shield bolt (2 required)
47	W714870W714870	Exhaust manifold nut — RH (8 required)

48	94309430	Exhaust manifold — RH
49	W714869W714869	Exhaust manifold stud — RH (8 required)
50	94489448	Exhaust manifold gasket — RH (2 required)
51	12K07312K073	Camshaft Position (CMP) sensor — LH
52	12K07312K073	CMP sensor bolt — LH
53	12K07312K073	CMP sensor bolt — RH
54	12K07312K073	CMP sensor — RH
55	60496049	Cylinder head — RH
56	1240512405	Upper spark plug (8 required)
57	1866318663	Heater coolant inlet fitting
58	W701571W701571	Heater coolant inlet fitting bolt
59	60656065	Cylinder head bolt — RH (10 required)
60	62506250	Camshaft — RH
61	W714203W714203	Camshaft bearing cap bolt — RH (10 required)
62	6B2846B284	Camshaft bearing cap — RH (4 required)
63	6B2806B280	Front camshaft bearing cap — RH
64	60526052	VCT system oil filter
65	6C5246C524	Camshaft phaser and sprocket — RH
66	62796279	Camshaft phaser and sprocket bolt — RH
67	6C5536C553	Exhaust rocker arm and shaft assembly — RH
68	6C5526C552	Intake rocker arm and shaft assembly — RH
69	65826582	Valve cover — RH
70	65826582	Oil fill cap
71	1228612286	Ignition wire — RH (4 required)
72	6A6666A666	PCV valve
73	6K8176K817	PCV tube
74	12C50812C508	Engine wiring harness

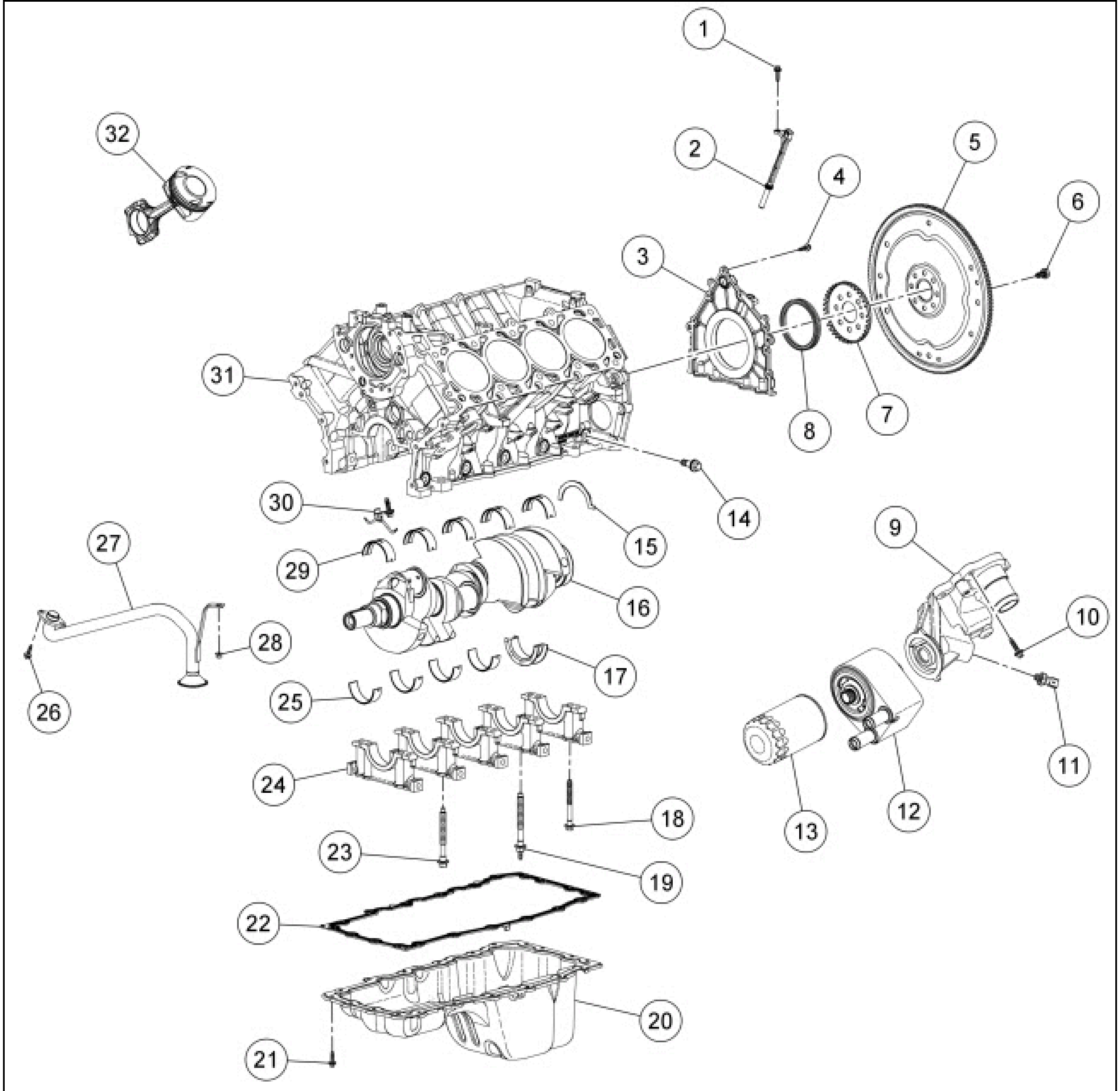
Engine — Front End



Item	Part Number	Description
1	6A3406A340	Crankshaft pulley bolt and washer
2	63166316	Crankshaft pulley
3	67006700	Crankshaft front seal
4	N808102N808102	Accessory drive belt idler pulley bolt
5	19A21619A216	Accessory drive belt idler pulley
6	N808102N808102	Accessory drive belt idler pulley bolt
7	19A21619A216	Accessory drive belt idler pulley
8	N806177N806177	Engine front cover bolt (17 required)
9	N804758N804758	Radio ignition interference capacitor nut
10	1880118801	Radio ignition interference capacitor
11	W713461W713461	Engine front cover stud bolt
12	W713462W713462	Engine front cover stud bolt (2 required)
13	6C0866C086	Engine front cover
14	6B2976B297	VCT variable force solenoid — RH
15	W503281W503281	Timing chain tensioner bolt — RH (2 required)
16	6L2666L266	Timing chain tensioner — RH
17	6B2976B297	VCT variable force solenoid — LH
18	6K2556K255	Timing chain tensioner arm — RH
19	62686268	Timing chain — RH
20	W503281W503281	Timing chain guide bolt — RH
21	6M2566M256	Timing chain guide — RH
22	6K2556K255	Tensioner arm — LH
23	W503281W503281	Timing chain tensioner bolt — LH (2 required)
24	6L2666L266	Timing chain tensioner — LH

25	62686268	Timing chain — LH
26	N606527N606527	Timing chain guide bolt — LH
27	6B2746B274	Timing chain guide — LH
28	W713655W713655	Oil pump stud bolt
29	63066306	Crankshaft sprocket
30	W713654W713654	Oil pump stud bolt
31	N806183N806183	Oil pump bolt (2 required)
32	66216621	Oil pump
33	N806282N806282	Coolant pump pulley bolt (4 required)
34	6A5286A528	Coolant pump pulley
35	W503298W503298	Coolant pump bolt (4 required)
36	85018501	Coolant pump
37	60106010	Cylinder block
38	W503276W503276	Heater coolant return fitting bolt
39	1866318663	Heater coolant return fitting
40	W503300W503300	Generator support bracket bolt (4 required)
41	1023910239	Generator support bracket
42	W500110W500110	Knock Sensor (KS) bolt (2 required)
43	12A69912A699	KS (2 required)
44	14B48514B485	KS wring harness

Engine — Lower End

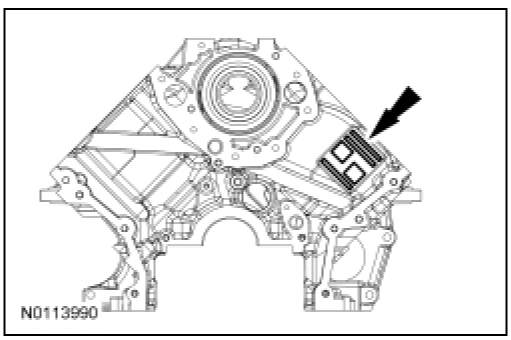


N0110910

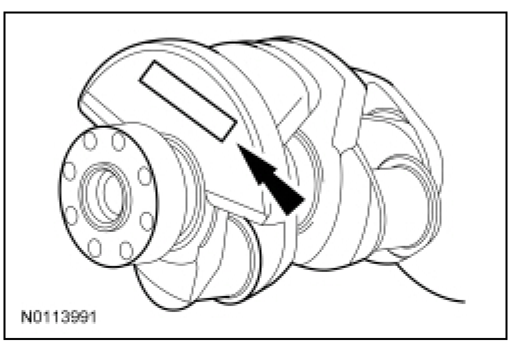
Item	Part Number	Description
1	W503276W503276	Crankshaft Position (CKP) sensor bolt
2	6C3156C315	CKP sensor
3	6K3186K318	Crankshaft rear seal retainer plate
4	N806155N806155	Crankshaft rear seal retainer plate bolt (10 required)
5	63756375	Flexplate
6	63796379	Flexplate bolt (8 required)
7	12A22712A227	Ignition pulse ring
8	67016701	Crankshaft rear seal
9	68816881	Oil filter adapter
10	W713261W713261	Oil filter adapter bolt (4 required)
11	92789278	Engine Oil Pressure (EOP) switch
12	6A6426A642	Oil cooler
13	67146714	Oil filter
14	6C3576C357	Crankshaft main bearing cap side bolt (10 required)

15	6A3416A341	Crankshaft thrust washer — upper
16	63036303	Crankshaft
17	6A3396A339	Crankshaft thrust washer — lower
18	63456345	Crankshaft main bearing cap bolt (10 required)
19	6K2586K258	Crankshaft main bearing cap stud bolt
20	66756675	Oil pan
21	W503276W503276	Oil pan bolt (20 required)
22	67106710	Oil pan gasket
23	63456345	Crankshaft main bearing cap bolt (9 required)
24	63256325	Crankshaft main bearing cap (5 required)
25	6A3386A338	Crankshaft bearing — lower (4 required)
26	W503276W503276	Oil pump screen and pickup tube bolt (2 required)
27	66226622	Oil pump screen and pickup tube
28	W701706W701706	Oil pump screen and pickup tube nut
29	63336333	Crankshaft bearing — upper (5 required)
30	6K8686K868	Piston oil cooler valve (4 required)
31	60106010	Cylinder block
32	—	Piston and connecting rod assembly (8 required)

1. Record the main bearing code found on the front of the engine block.



2. Record the main bearing code found on the back of the crankshaft.



3. Using the data recorded earlier and the Bearing Select Fit Chart, Standard Bearings, determine the required bearing grade for each main bearing.
    - Read the first letter of the engine block main bearing code and the first letter of the crankshaft main bearing code.
    - Read down the column below the engine block main bearing code letter, and across the row next to the crankshaft main bearing code letter, until the 2 intersect. This is the required bearing grade for the No. 1

crankshaft main bearing.

- As an example, if the engine block code letter is F and the crankshaft code letter is D, the correct bearing grade for this main bearing is a 2.
- Repeat this process for the remaining 4 main bearings.

MINIMUM BLOCK DIA

MAXIMUM CRANKSHAFT DIA

	72.400	A .401	B .402	C .403	D .404	E .405	F .406	G .407	H .408	I .409	J .410	K .411	L .412	M .413	N .414	O .415	P .416	Q .417	R .418	S .419	T .420	U .421	V .422	W .423	X .424
X	67.504	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
W	67.503	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
V	67.502	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2
U	67.501	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2
T	67.500	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2
S	67.499	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2
R	67.498	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Q	67.497	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
P	67.496	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
O	67.495	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3
N	67.494	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3
M	67.493	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3
L	67.492	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3
K	67.491	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3
J	67.490	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3
I	67.489	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3
H	67.488	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3
G	67.487	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3
F	67.486	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3
E	67.485	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3
D	67.484	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3
C	67.483	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3
B	67.482	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
A	67.481	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

N0113992

4. If oversize bearings are being used, use the procedure in the previous step and the Bearing Select Fit Chart, Oversize Bearings to determine the required bearing grade for each main bearing.

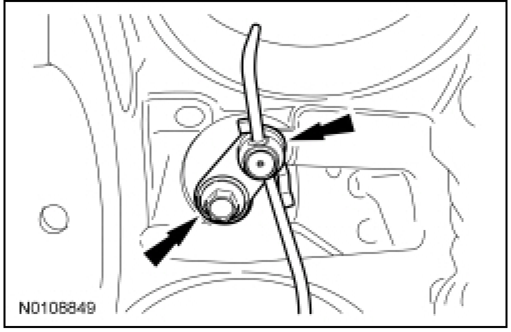
MINIMUM BLOCK DIA

MAXIMUM CRANKSHAFT DIA

	72.400	A .401	B .402	C .403	D .404	E .405	F .406	G .407	H .408	I .409	J .410	K .411	L .412	M .413	N .414	O .415	P .416	Q .417	R .418	S .419	T .420	U .421	V .422	W .423	X .424
X	67.254	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
W	67.253	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
V	67.252	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2
U	67.251	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2
T	67.250	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2
S	67.249	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2
R	67.248	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Q	67.247	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
P	67.246	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
O	67.245	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3
N	67.244	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3
M	67.243	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3
L	67.242	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3
K	67.241	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3
J	67.240	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3
I	67.239	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3
H	67.238	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3
G	67.237	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3
F	67.236	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3
E	67.235	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3
D	67.234	2	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3
C	67.233	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3
B	67.232	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
A	67.231	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

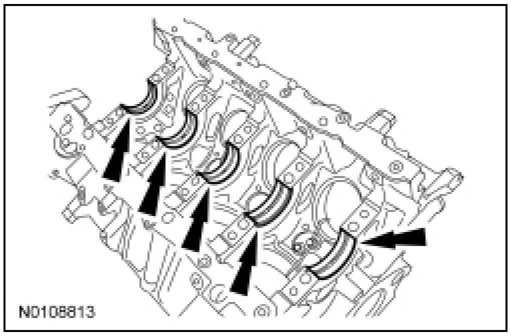
N0113993

5. Position the 4 piston oil cooler valves with the bolt boss facing toward the rear of the cylinder block and install the 4 bolts in 2 stages.
   - Stage 1: Tighten to 20 Nm (177 lb-in).
   - Stage 2: Tighten an additional 60 degrees.



6. **NOTE:** *Before assembling the cylinder block, all sealing surfaces must be free of chips, dirt, paint and foreign material. Also, make sure the coolant and oil passages are clear.*

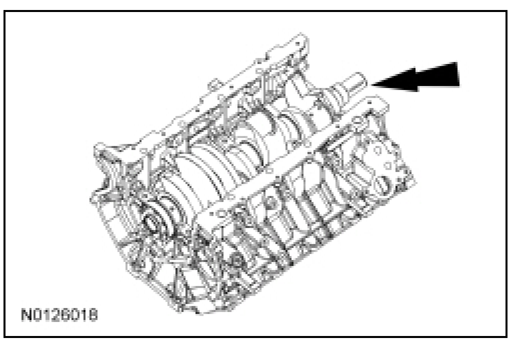
   Install the crankshaft upper main bearings into the cylinder block and lubricate them with clean engine oil.



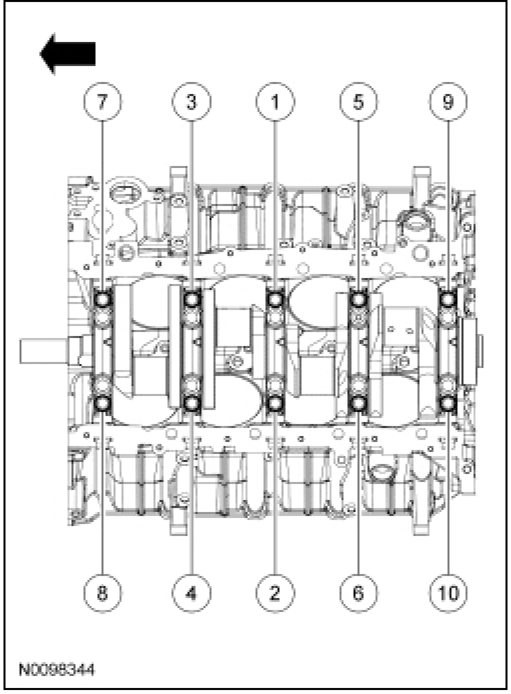
7. **NOTICE: Do not allow the crankshaft to come into contact with the piston oil cooler valves during assembly or engine damage may occur.**

   NOTE: *The upper thrust washers are shown for location purposes only. Do not install the upper thrust washers until the crankshaft is installed. Refer to the following 2 steps.*

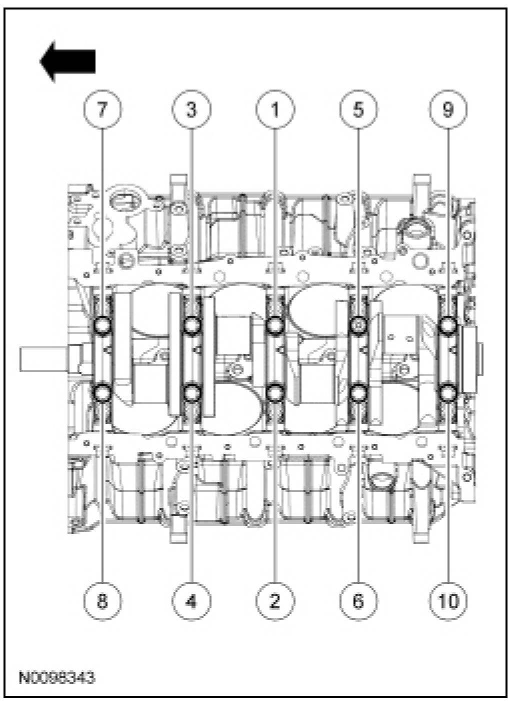
   Install the crankshaft onto the upper crankshaft main bearings.



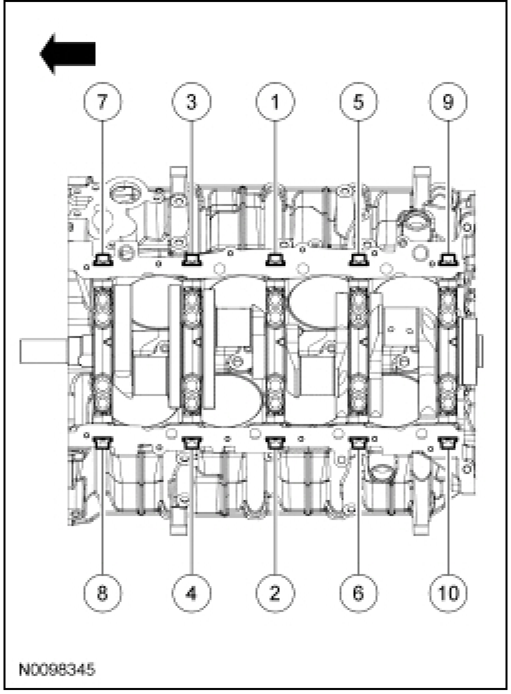
8. Install the lower crankshaft flange bearing onto the No. 5 main bearing cap, with oil grooves facing the crankshaft surface.
9. Install the crankshaft lower main bearings into the main bearing caps and lubricate them with clean engine oil. Locate the main bearing cap on the cylinder block and, keeping the cap as square as possible, alternately draw the cap down evenly using the cap fasteners.
10. Install the 20 vertical inner and outer main bearing cap fasteners and the main bearing side bolts finger-tight.
11. Apply a forward load to the crankshaft, so that the crankshaft thrust face is seated.
12. Tighten the 10 vertical outer main bearing cap fasteners in 3 stages, in the sequence shown.
    - Stage 1: Tighten to 35 Nm (26 lb-ft).
    - Stage 2: Tighten to 50 Nm (26 lb-ft).
    - Stage 3: Tighten an additional 90 degrees.



13. Tighten the 10 vertical inner main bearing cap fasteners in 3 stages, in the sequence shown.
    - Stage 1: Tighten to 35 Nm (26 lb-ft).
    - Stage 2: Tighten to 65 Nm (48 lb-ft).
    - Stage 3: Tighten an additional 90 degrees.

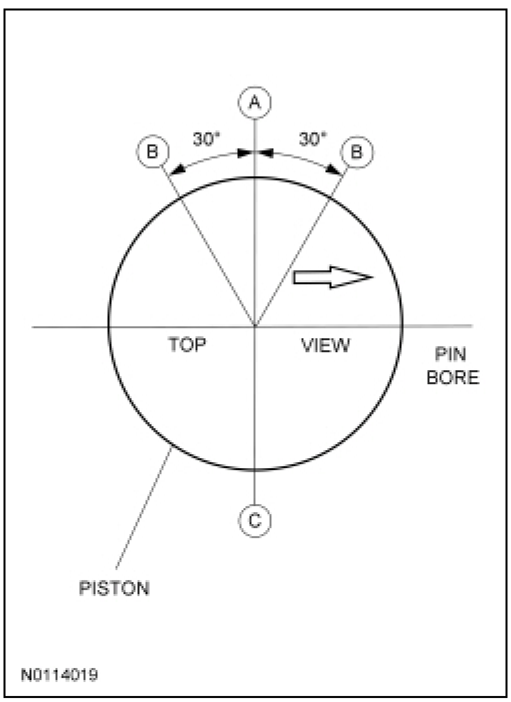


14. Install the 10 side bolts and tighten them in 3 stages, in the sequence shown.
    - Stage 1: Tighten to 20 Nm (177 lb-in).
    - Stage 2: Tighten to 35 Nm (26 lb-ft).
    - Stage 3: Tighten an additional 60 degrees.



N0098345

15. Check the crankshaft end play clearance. For additional information, refer to Section 303-00.
16. Assemble the 8 pistons. For additional information, refer to Piston in this section.
17. Make sure the ring gaps (upper compression ring — A, oil control segment ring — B, expander ring and lower compression ring — C) are correctly spaced around the circumference of the piston.



N0114019

18. **NOTICE: Do not allow the connecting rod to come into contact with the piston oil cooler valves during assembly or engine damage may occur.**

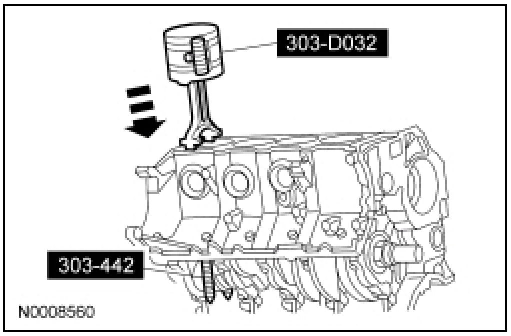
    NOTICE: Do not scratch the cylinder walls or crankshaft journals with the connecting rod or engine damage may occur.

    NOTE: *The following piston installation steps are for all 8 connecting rods, rod bearings and pistons. Only one connecting rod, rod bearing and piston is shown.*

    NOTE: *The arrow on the top of the piston faces the front of the engine.*

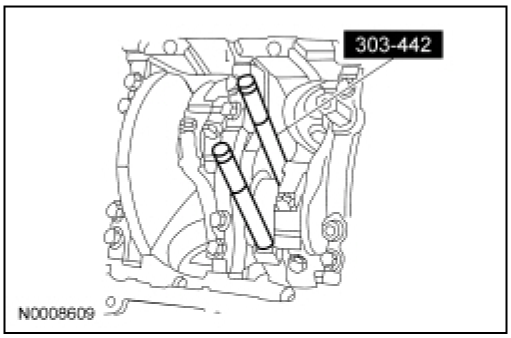
Using the Piston Ring Compressor and the Connecting Rod Installer, install the connecting rod with the upper connecting rod bearing in place.

- Lubricate the piston and ring with clean engine oil.
- Lubricate the rod bearings with clean engine oil.



19. **NOTICE: Do not scratch the cylinder walls or crankshaft journals with the connecting rod or engine damage may occur.**

    Once the connecting rod is seated on the crankshaft journal, remove the Connecting Rod Installer.



20. **NOTICE: The rod cap installation must keep the same orientation as marked during disassembly or engine damage may occur.**

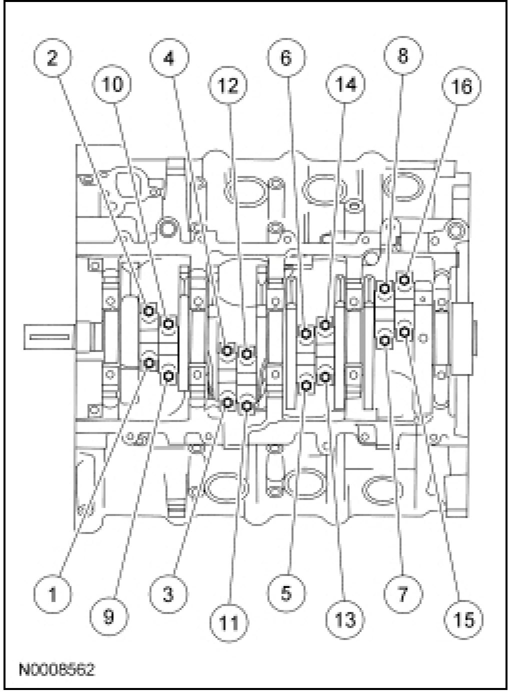
    NOTE: *The connecting rod caps are of the "cracked" design and must mate with the connecting rod ends. Excessive bearing clearance will result if not mated correctly.*

    Position the lower bearing and connecting rod, and install the 16 new bolts loosely.

21. Check the piston-to-cylinder block and piston ring clearances. For additional information, refer to Section 303-00.

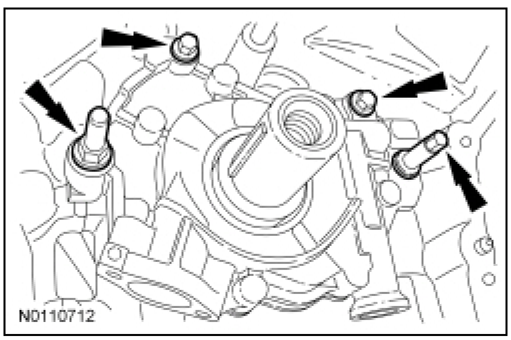
22. **NOTE:** *Main bearing caps are removed for clarity.*

    Tighten the 16 bolts in 2 stages, in the sequence shown.
    - Stage 1: Tighten to 20 Nm (177 lb-in).
    - Stage 2: Tighten to 43 Nm (32 lb-ft).
    - Stage 3: Tighten an additional 135 degrees.



23. Prime the oil pump. Add 2 tablespoons of clean engine oil to the oil pump and rotate the oil pump by hand.

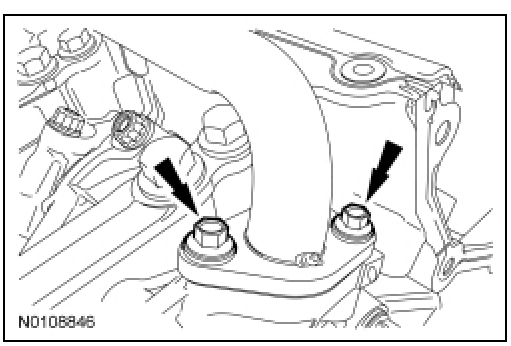
24. Position the oil pump and install the 2 upper bolts and the lower stud bolts in 5 stages.
    - Stage 1: Tighten all fasteners to 2 Nm (18 lb-in).
    - Stage 2: Tighten the 2 upper bolts to 10 Nm (89 lb-in).
    - Stage 3: Tighten the 2 lower stud bolts to 20 Nm (177 lb-in).
    - Stage 4: Tighten the 2 upper bolts an additional 45 degrees.
    - Stage 5: Tighten the 2 lower stud bolts an additional 60 degrees.



25. **NOTICE: Make sure the O-ring is in place and not damaged. A missing or damaged O-ring can cause foam in the lubrication system, low oil pressure and severe engine damage.**

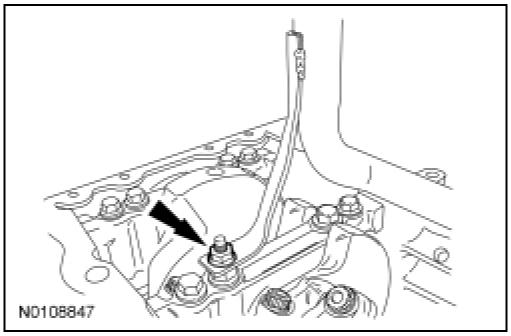
    NOTE: *Clean and inspect the mating surfaces and install a new O-ring. Lubricate the O-ring with clean engine oil prior to installation.*

    Position the oil pump screen and pickup tube and install the 2 front bolts.
    - Tighten to 10 Nm (89 lb-in).



26. Install the oil pump screen and pickup tube support bracket nut.

- Tighten to 24 Nm (18 lb-ft).



27. Position the crankshaft keyway at the 11 o'clock position.
28. **NOTICE: Make sure all coolant residue and foreign material are cleaned from the block surface and cylinder bore.**

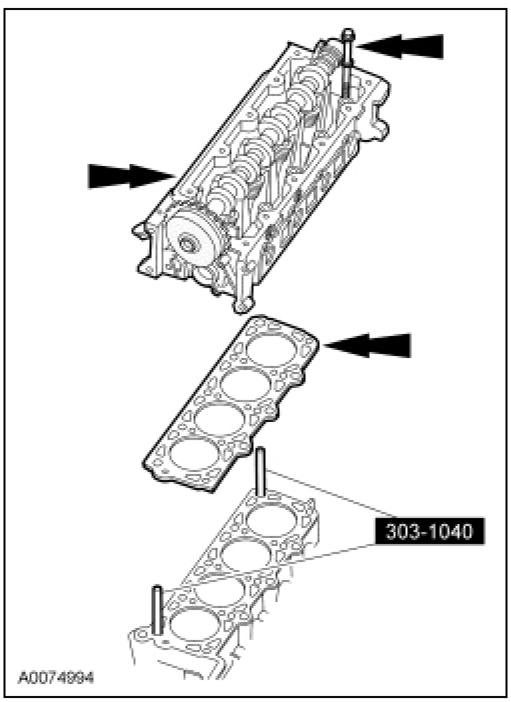
    NOTICE: The use of sealing aids (aviation cement, copper spray and glue) is not permitted. The gasket must be installed dry or engine damage may occur.

    NOTICE: The cylinder head bolts must be discarded and new bolts installed. They are a tighten-to-yield design and cannot be reused.

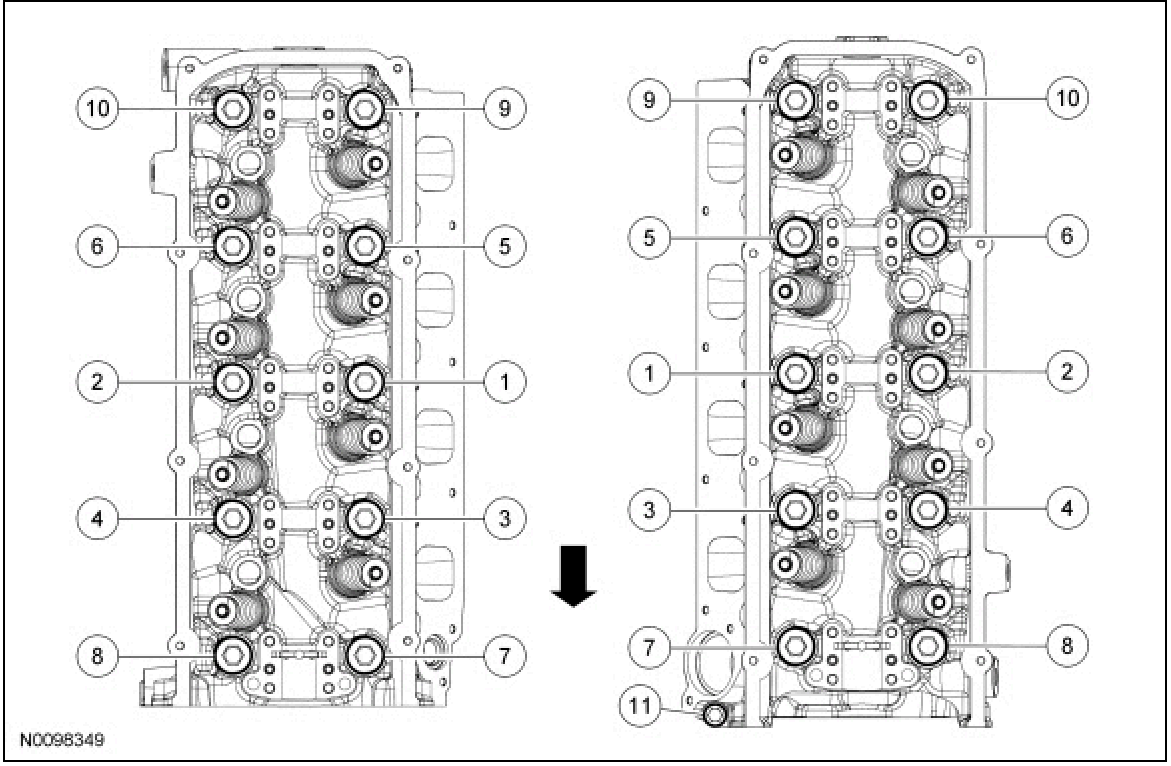
    NOTE: *Do not turn the crankshaft until instructed to do so.*

    NOTE: *LH shown, RH similar.*

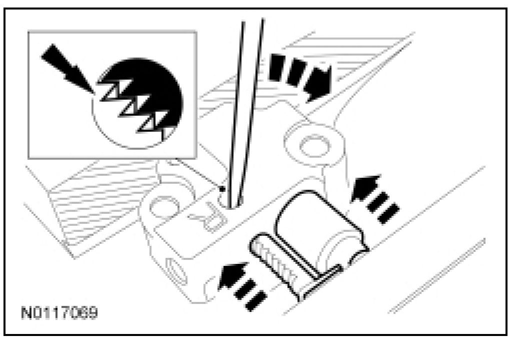
    Using the Cylinder Head Alignment Pins, position the new cylinder head gaskets and cylinder heads over the dowels and install the 20 cylinder head bolts loosely.



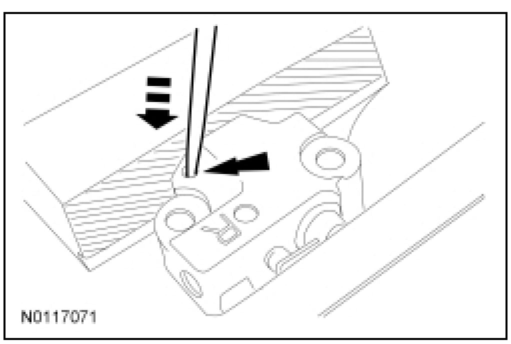
29. Tighten the 21 bolts in 6 stages, in the sequence shown.
    - Stage 1: Tighten the M12 fasteners to 25 Nm (18 lb-ft).
    - Stage 2: Tighten the M12 fasteners to 60 Nm (44 lb-ft).
    - Stage 3: Tighten the M12 fasteners an additional 90 degrees.
    - Stage 4: Tighten the M12 fasteners an additional 90 degrees.
    - Stage 5: Tighten the M8 fastener to 20 Nm (18 lb-ft).
    - Stage 6: Tighten the M8 fastener and additional 45 degrees.



30. Using a small pick, carefully push the tensioner rack pawl retainer away from the rack pawl and compress the tensioner plunger and rack using a vise.

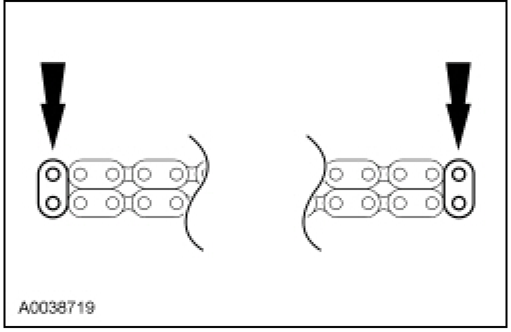


31. Install a small pick into the tensioner to hold the rack pawl and plunger in the seated position for tensioner installation.



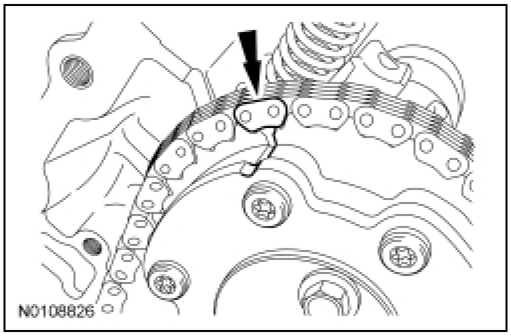
32. Remove the tensioner from the vise.

33. If the blue links are not visible, mark 2 links on each end, and use as timing marks.

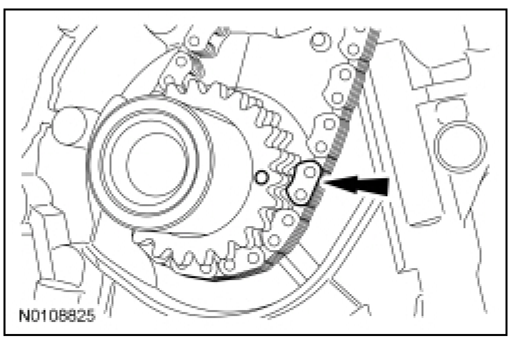


34. **NOTE:** *Make sure the upper half of the timing chain is below the tensioner arm dowel and above the chain guide pin.*

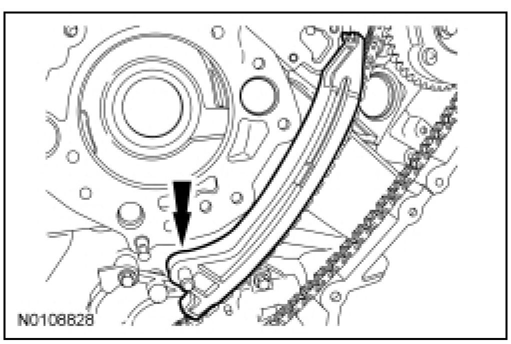
    Position the upper end of the LH (inner) timing chain on the LH camshaft phaser and sprocket, aligning the timing mark on the outer flange of the camshaft phaser and sprocket with the single blue (marked) link on the chain.



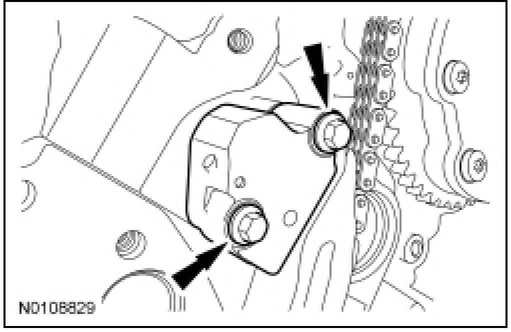
35. Position the lower end of the LH (inner) timing chain on the crankshaft sprocket, aligning the timing mark on the outer flange of the crankshaft sprocket with the single blue (marked) link on the chain and install the crankshaft sprocket onto the crankshaft and verify the crankshaft keyway is at the 11 o'clock position.



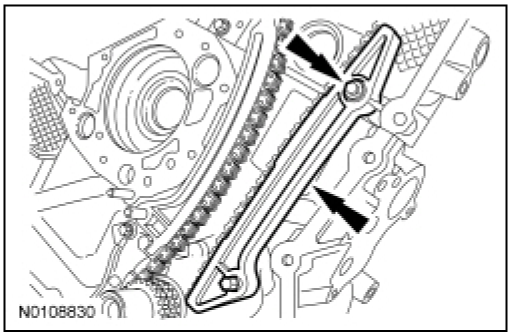
36. Install the LH timing chain tensioner arm.



37. Position the LH timing chain tensioner and install the 2 bolts in 2 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten an additional 45 degrees.

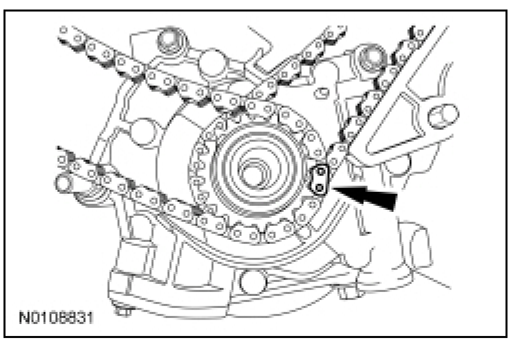


38. Position the LH timing chain guide and install the bolt in 2 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten an additional 45 degrees.

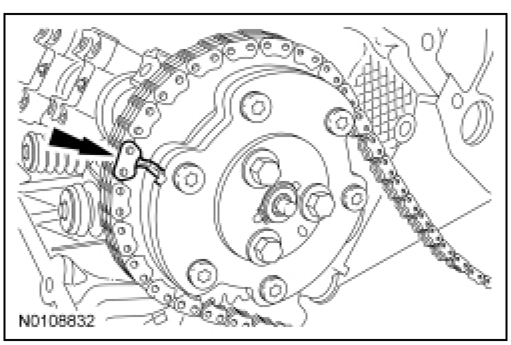


39. Remove the retaining clip from the LH timing chain tensioner.

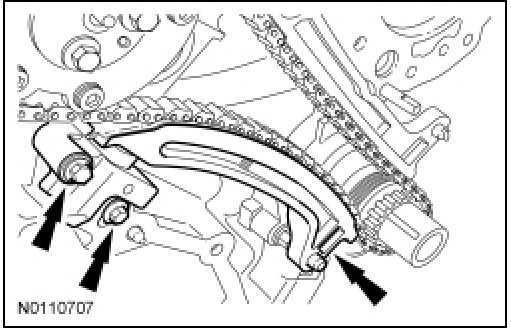
40. Position the lower end of the RH (outer) timing chain on the crankshaft sprocket, aligning the timing mark on the sprocket with the single blue (marked) chain link.



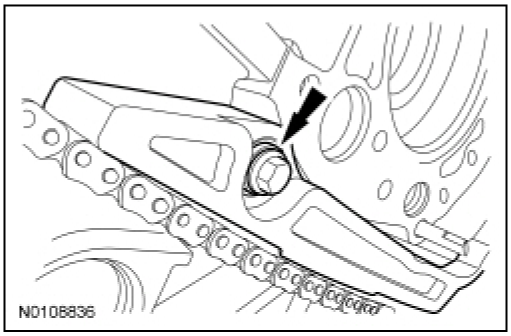
41. Position the upper end of the RH (outer) timing chain on the RH camshaft phaser and sprocket, aligning the timing mark on the outer flange of the camshaft phaser and sprocket with the single blue (marked) link on the chain.



42. Position the RH timing chain tensioner arm on the dowel pin and the RH timing chain tensioner and install the 2 bolts in 2 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten an additional 45 degrees.

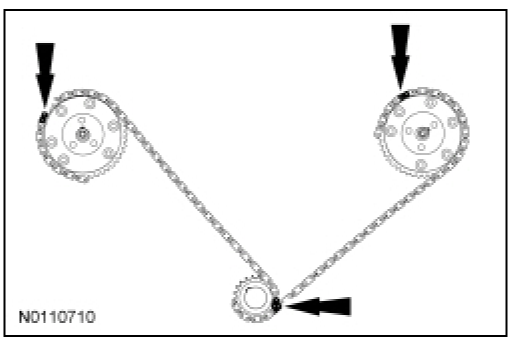


43. Position the RH timing chain guide and install the bolt in 2 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten an additional 45 degrees.



44. Remove the retaining clip from the RH timing chain tensioner.

45. As a post-check, verify correct alignment of all timing marks.

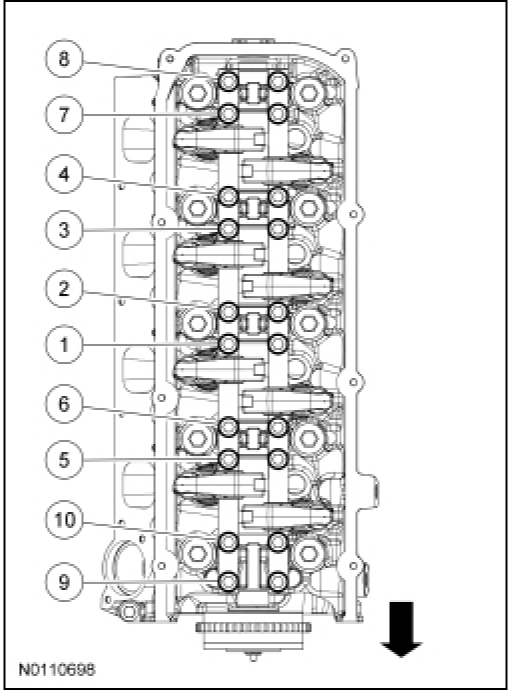


46. Lubricate the rocker arm shaft-to-valve tip and rocker arm-to-camshaft lobe contact area with clean engine oil prior to installation.

47. Rotate the engine clockwise until the No. 1 cylinder camshaft intake lobe is positioned so that the intake valve would be fully opened.

48. Position the LH intake rocker arm shaft assembly and install the bolts finger-tight.

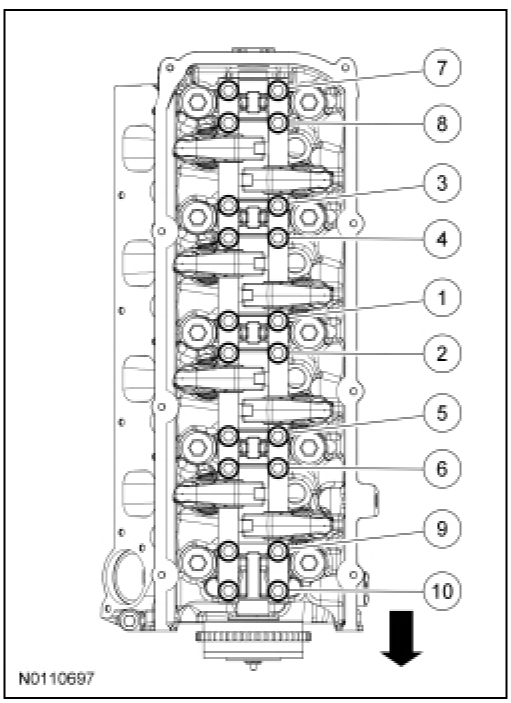
49. Tighten the 10 LH intake rocker arm shaft assembly bolts in the sequence shown, in 3 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten to 20 Nm (177 lb-in).
    - Stage 3: Tighten an additional 60 degrees.



50. Rotate the engine clockwise until the No. 1 cylinder camshaft exhaust lobe is positioned so that the exhaust valve would be fully opened.

51. Position the LH exhaust rocker arm shaft assembly and install the bolts finger-tight.

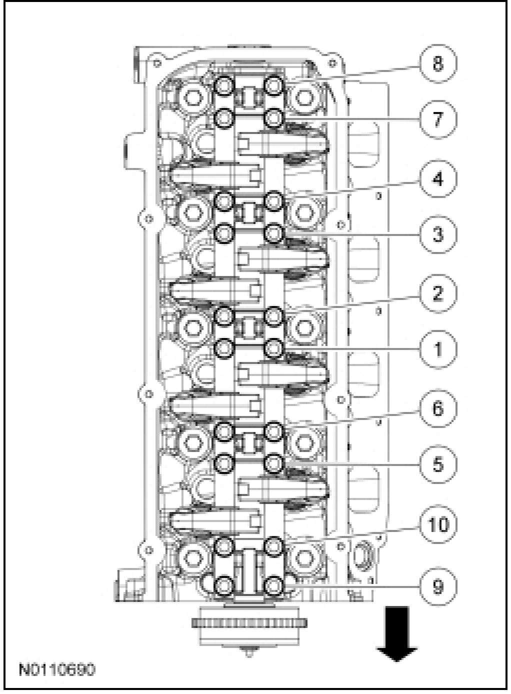
52. Tighten the 10 LH exhaust rocker arm shaft assembly bolts in the sequence shown, in 3 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten to 20 Nm (177 lb-in).
    - Stage 3: Tighten an additional 60 degrees.



53. Rotate the engine clockwise until the No. 1 cylinder camshaft intake lobe is positioned so that the intake valve would be fully closed.

54. Position the RH intake rocker arm shaft assembly and install the bolts finger-tight.

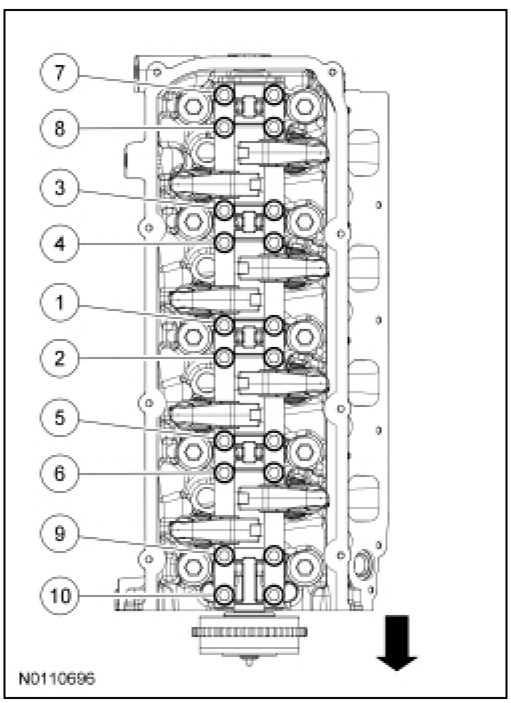
55. Tighten the 10 RH intake rocker arm shaft assembly bolts in the sequence shown, in 3 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten to 20 Nm (177 lb-in).
    - Stage 3: Tighten an additional 60 degrees.



56. Rotate the engine clockwise until the No. 1 cylinder camshaft exhaust lobe is positioned so that the exhaust valve would be fully closed.

57. Position the RH exhaust rocker arm shaft assembly and install the bolts finger-tight.

58. Tighten the 10 RH exhaust rocker arm shaft assembly bolts in the sequence shown, in 3 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten to 20 Nm (177 lb-in).
    - Stage 3: Tighten an additional 60 degrees.

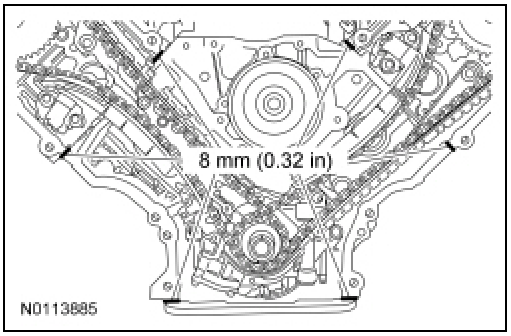


59. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

    NOTE: *If the engine front cover is not secured within 6 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Follow the directions on the packaging. Failure to follow this procedure can cause future oil leakage.*

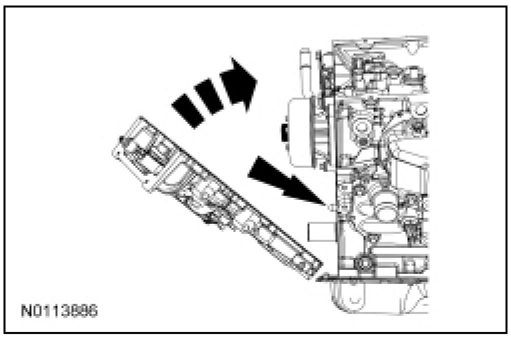
NOTE: *Make sure that the engine front cover gasket is in place on the engine front cover before installation.*

Apply a bead of silicone gasket and sealant along the cylinder head-to-cylinder block surface and the oil pan-to-cylinder block surface, at the locations shown.



60. **NOTICE: The Variable Camshaft Timing (VCT) variable force solenoid pins must be fully depressed to avoid interference with the VCT valve tips when installing the engine front cover. Failure to follow these instructions can result in damage to the engine.**

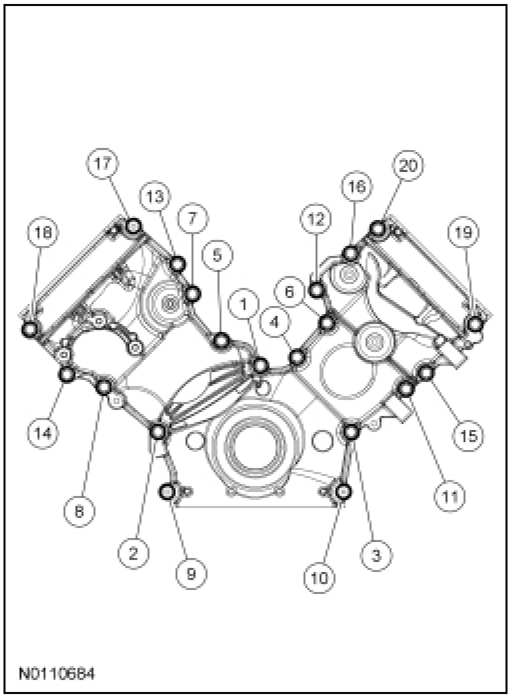
    Install new engine front cover gaskets on the engine front cover. With the VCT variable force solenoid pins fully depressed, position the engine front cover onto the dowels. Install the fasteners finger-tight.



61. Tighten the 20 engine front cover fasteners in the sequence shown in 3 stages.
    Stage 1: Tighten all fasteners to 10 Nm (89 lb-in).
    Stage 2: Tighten all fasteners to 20 Nm (177 lb-in).
    Stage 3: Tighten all fasteners an additional 45 degrees.

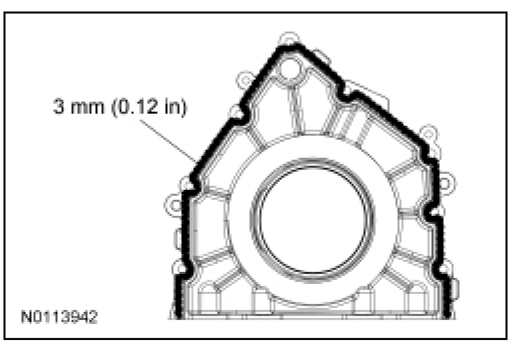
Item	Part Number	Description
1	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
2	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
3	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
4	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
5	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
6	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
7	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
8	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
9	W713462	Stud, Hex Head Pilot, M8 x 33 + M10 x 30
10	W713462	Stud, Hex Head Pilot, M8 x 33 + M10 x 30
11	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
12	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
13	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33

Item	Part Number	Description
14	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
15	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
16	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
17	W713461	Stud, Hex Head Pilot, M8 x 33 + M8 x 27
18	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
19	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
20	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33

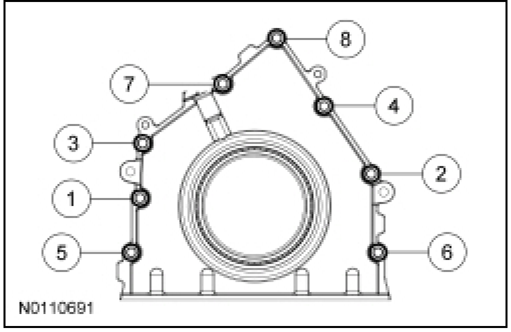


62. **NOTE:** *If the rear crankshaft seal retaining plate is not secured within 6 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure may cause future oil leaks.*

    Apply a bead of silicone gasket and sealant around the groove along the rear of crankshaft rear seal retainer plate sealing surface.

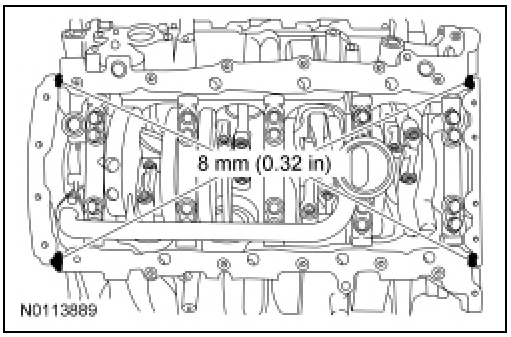


63. Install the crankshaft rear seal retainer plate and the 8 bolts in the sequence shown, in 2 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten an additional 45 degrees.

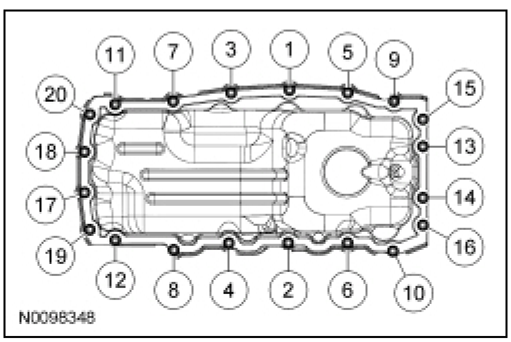


64. **NOTE:** *If not secured within 6 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.*

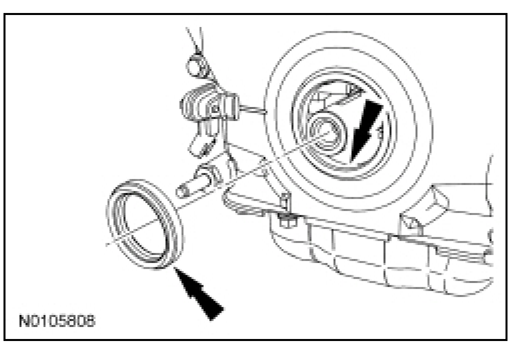
    Apply silicone gasket and sealant at the crankshaft rear seal retainer plate-to-cylinder block sealing surface and at the engine front cover-to-cylinder block sealing surface.



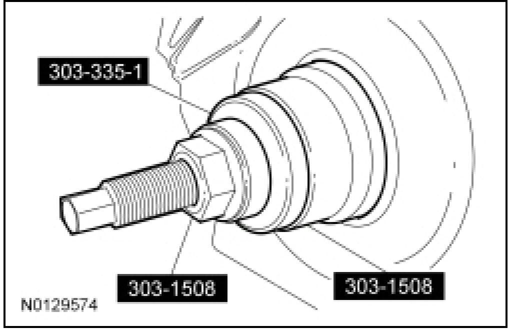
65. Position a new gasket and the oil pan and install the 20 bolts.
    - Tighten the bolts in the sequence shown in 3 stages.
        - Stage 1: Tighten to 2 Nm (18 lb-in).
        - Stage 2: Tighten to 10 Nm (89 lb-in).
        - Stage 3: Tighten an additional 45 degrees.



66. Lubricate the engine front cover and the new crankshaft seal inner lip with clean engine oil.

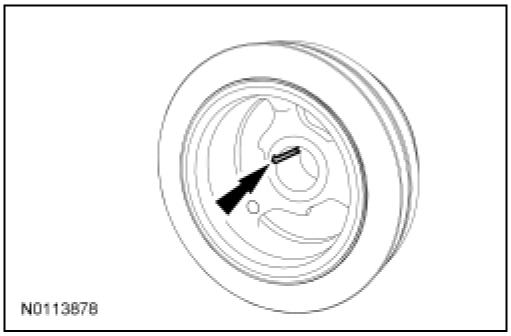


67. Using the Crankshaft Damper and Crankshaft Front Seal Installer and the Front Cover Oil Seal, install the new crankshaft front seal into the engine front cover.

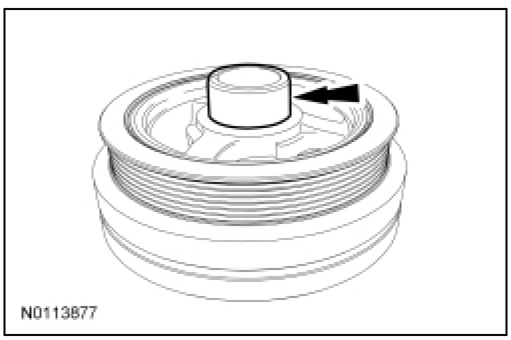


68. **NOTE:** *If not secured within 6 minutes, the sealant must be removed and the sealing area cleaned with metal surface prep and silicone gasket remover. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.*

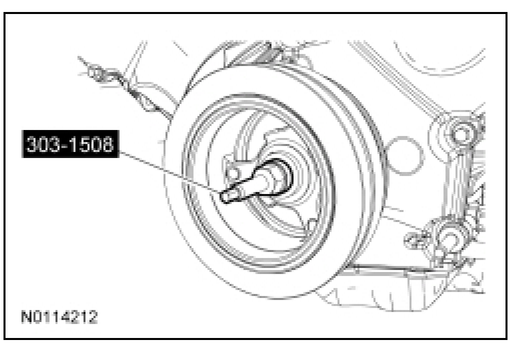
    Apply silicone gasket and sealant to the Woodruff key slot in the crankshaft pulley.



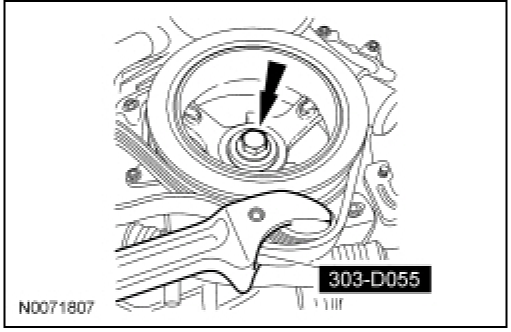
69. Lubricate the crankshaft pulley sealing area with clean engine oil prior to installation.



70. Using the Crankshaft Vibration Damper Installer, install the crankshaft pulley.



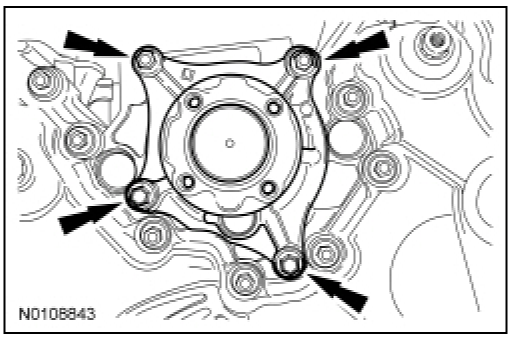
71. Using the Strap Wrench, install a new crankshaft pulley bolt and the original washer, tighten the bolt in 2 stages.
    - Stage 1: Tighten to 175 Nm (129 lb-ft).
    - Stage 2: Tighten an additional 90 degrees.



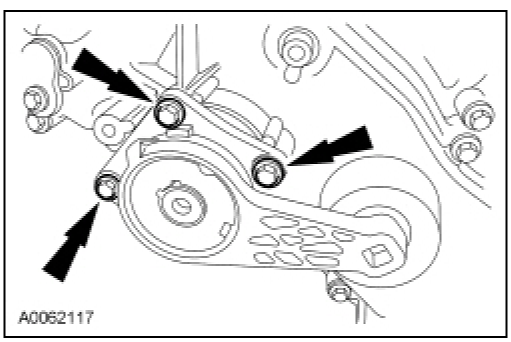
72. **NOTICE: Do not rotate the coolant pump housing once the coolant pump has been positioned in the cylinder block. Damage to the O-ring seal will occur.**

    NOTE: *Lubricate the new O-ring seal using clean engine coolant prior to installation.*

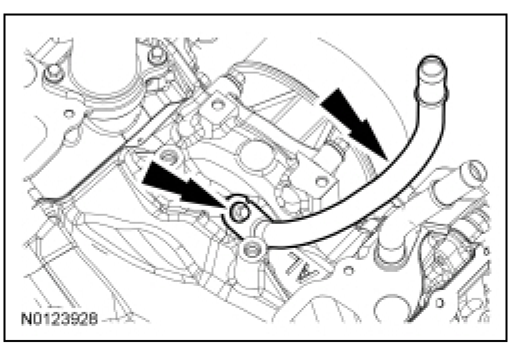
    Using a new O-ring seal, position the coolant pump and install the 4 bolts in 2 stages.
    - Stage 1: Tighten to 20 Nm (117 lb-in).
    - Stage 2: Tighten an additional 45 degrees.



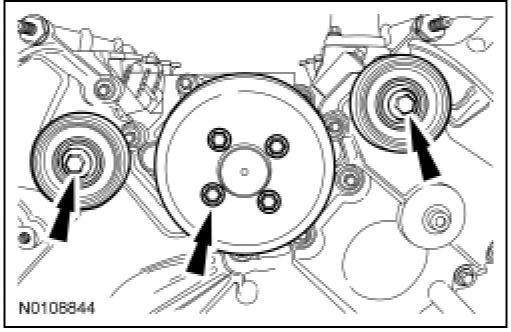
73. Install the accessory drive belt tensioner and the 3 bolts.
    - Tighten to 25 Nm (18 lb-ft).



74. Install the heater coolant tube and the bolt.
    - Tighten to 10 Nm (89 lb-in).



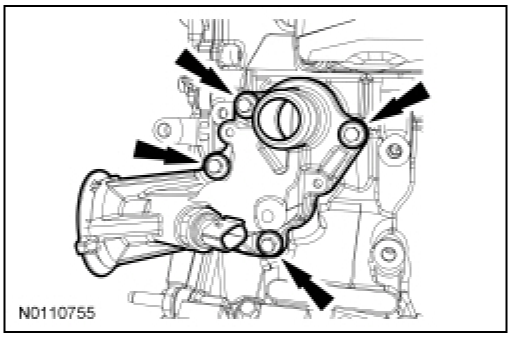
75. Install the 2 accessory drive idler pulleys, the coolant pump pulley and the 6 bolts.
    - Tighten to 25 Nm (18 lb-ft).



76. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

    NOTE: *Clean and inspect the mating surfaces, and install a new gasket.*

    Position the oil filter adapter and install the 4 bolts.
    - Tighten to 25 Nm (18 lb-ft).

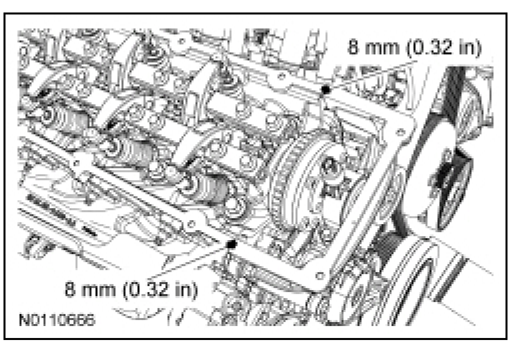


77. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

    Clean the valve cover mating surface with silicone gasket remover and metal surface prep. Follow the directions on the packaging.

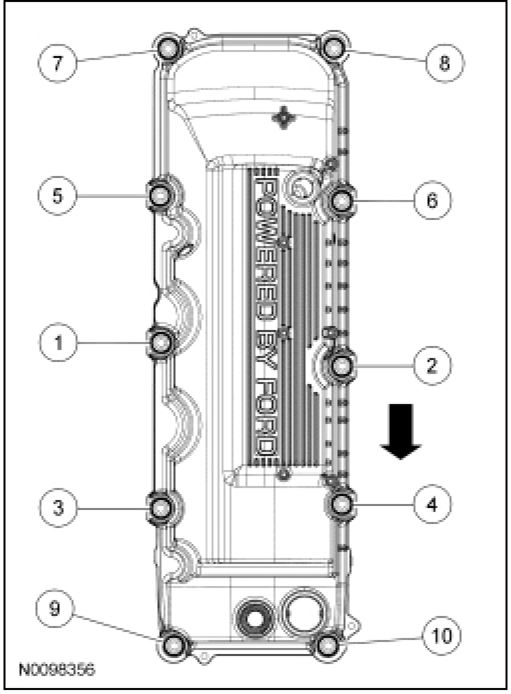
78. **NOTE:** *If the valve cover is not secured within 6 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.*

    Apply a bead of silicone gasket and sealant in 2 places where the engine front cover meets the cylinder head.



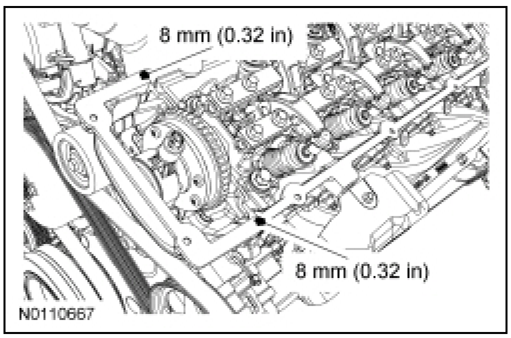
79. **NOTICE: When installing the valve cover, make sure to avoid damaging the Variable Camshaft Timing (VCT) variable force solenoid.**

    Position the RH valve cover and new gasket on the cylinder head and tighten the 10 fasteners in the sequence shown.
    - Tighten to 10 Nm (89 lb-in).



80. **NOTE:** *If the valve cover is not secured within 6 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.*

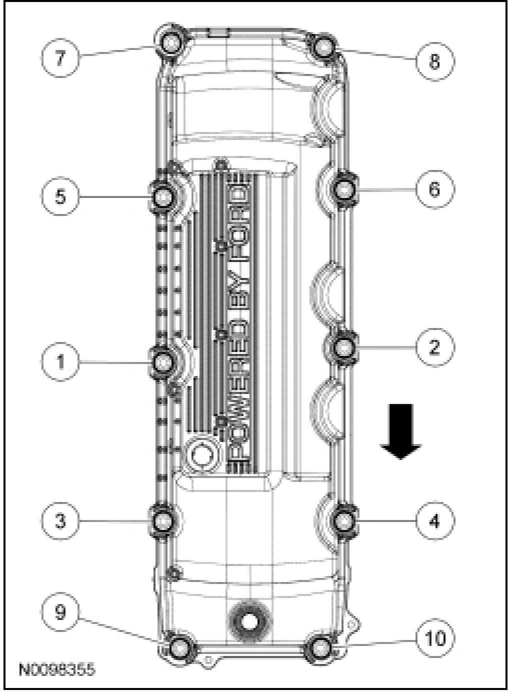
    Apply a bead of silicone gasket and sealant in 2 places where the engine front cover meets the cylinder head.



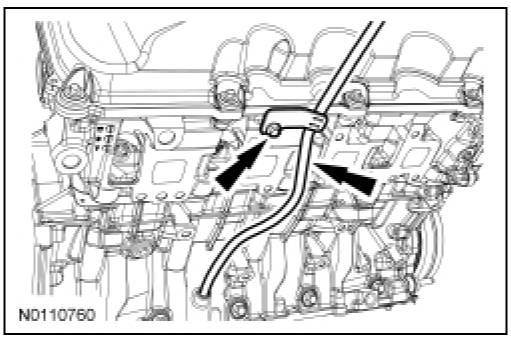
81. **NOTICE: When installing the valve cover, make sure to avoid damaging the Variable Camshaft Timing (VCT) variable force solenoid.**

    Position the LH valve cover and new gasket on the cylinder head and tighten the 10 fasteners in the sequence shown.

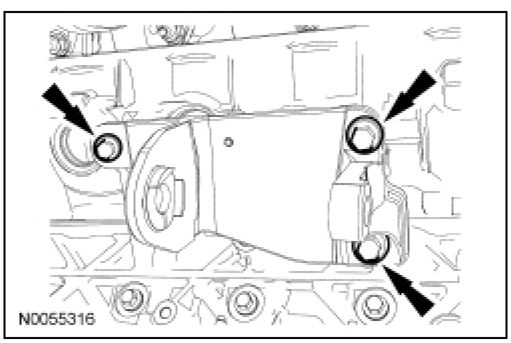
    - Tighten to 10 Nm (89 lb-in).



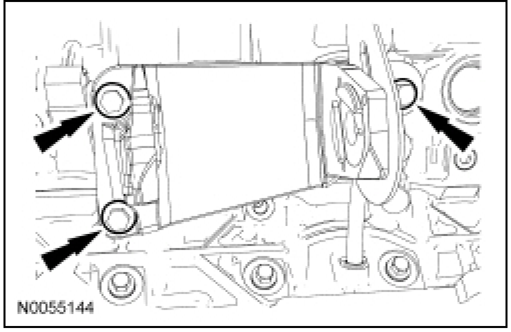
82. Install the oil level indicator tube and the bolt.
    - Install a new O-ring seal and lubricate the O-ring seal with clean engine oil prior to installation.
    - Tighten to 10 Nm (89 lb-in).



83. Position the RH engine support insulator-to-cylinder block bracket and install the 3 bolts.
    - Tighten to 63 Nm (46 lb-ft).

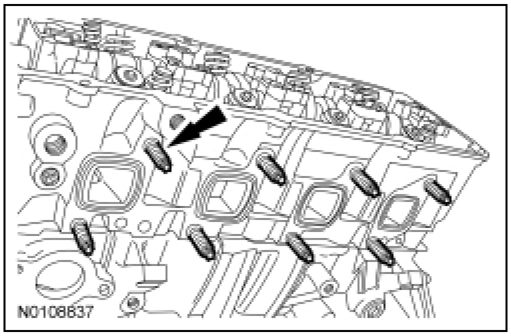


84. Position the RH engine support insulator-to-cylinder block bracket and install the 3 bolts.
    - Tighten to 63 Nm (46 lb-ft).

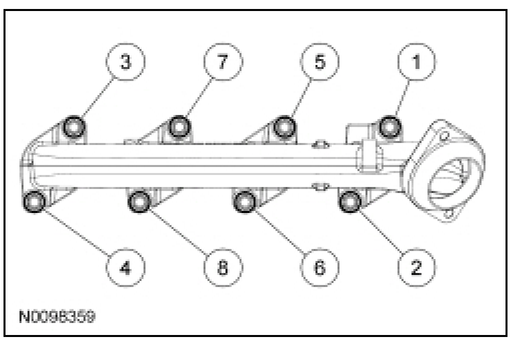


85. **NOTE:** *LH shown, RH similar.*

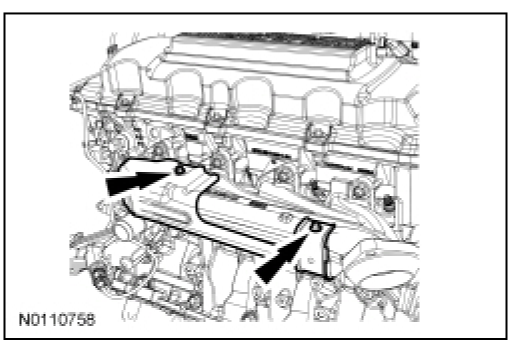
    Install 16 new exhaust manifold studs.
    - Tighten to 25 Nm (18 lb-ft).



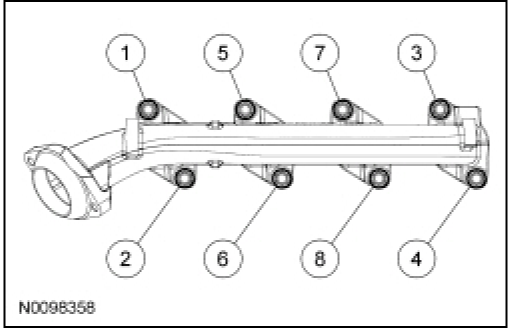
86. Using new exhaust manifold nuts, position the LH exhaust manifold and install the 8 nuts in 2 stages in the sequence shown.
    - Stage 1: Tighten to 25 Nm (18 lb-ft).
    - Stage 2: Tighten to 32 Nm (24 lb-ft).



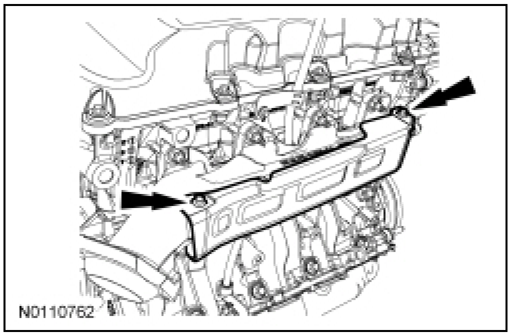
87. Position the LH exhaust manifold heat shield and install the 2 bolts.
    - Tighten to 12 Nm (106 lb-in).



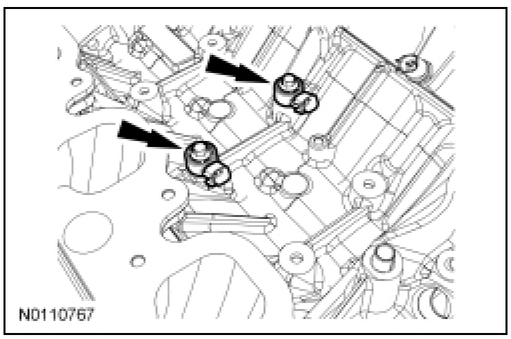
88. Using new exhaust manifold nuts, position the LH exhaust manifold and install the 8 nuts in 2 stages in the sequence shown.
    - Stage 1: Tighten to 25 Nm (18 lb-ft).
    - Stage 2: Tighten to 32 Nm (24 lb-ft).



89. Position the RH exhaust manifold heat shield and install the 2 bolts.
    - Tighten to 12 Nm (106 lb-in).

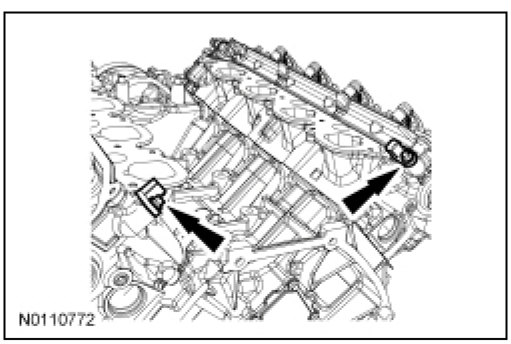


90. Install the 2 Knock Sensor (KS) and the 2 bolts.
    - Tighten to 20 Nm (177 lb-in).



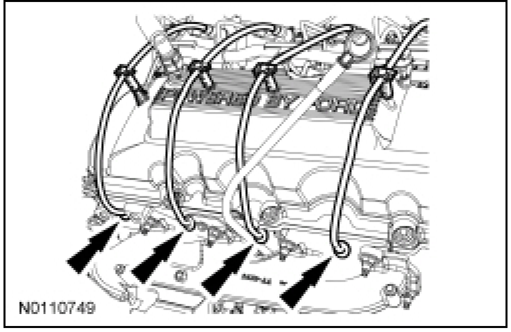
91. **NOTE:** *Lubricate the O-ring seal with clean engine oil prior to installation.*

    Install the LH and RH Camshaft Position (CMP) sensors and the 2 bolt.
    - Tighten to 10 Nm (89 lb-in).



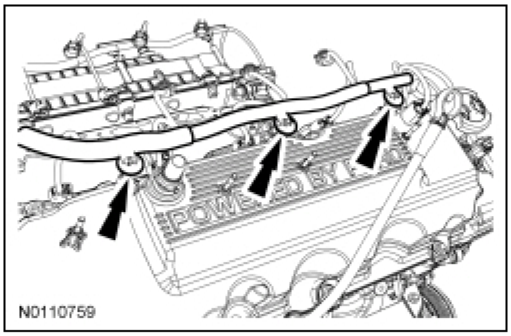
92. **NOTE:** *RH shown, LH similar.*

    Connect the 8 ignition wires to the valve cover retainers and the lower spark plugs.

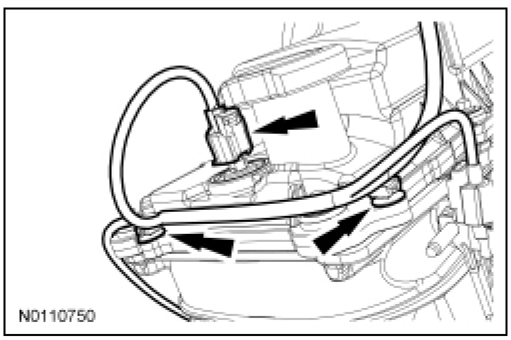


93. Position the wiring harness onto the engine assembly.

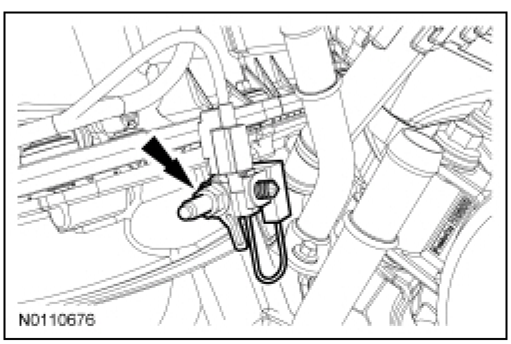
94. Connect the 3 engine wiring harness position retainers to the RH valve cover.



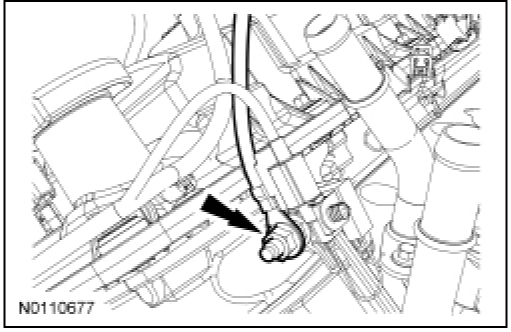
95. Connect the RH VCT variable force solenoid electrical connector and the 2 wiring harness retainers.



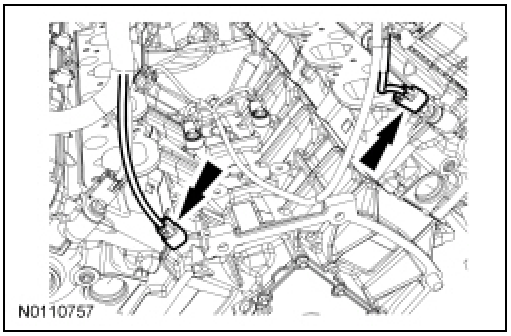
96. Position the radio ignition interference capacitor and install the nut.
   - Tighten to 25 Nm (18 lb-ft).



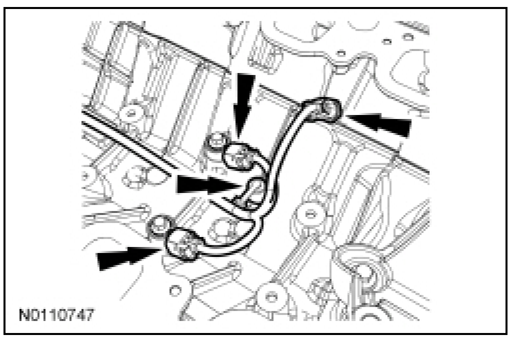
97. Position the ground cable and install the nut.
   - Tighten to 25 Nm (18 lb-ft).



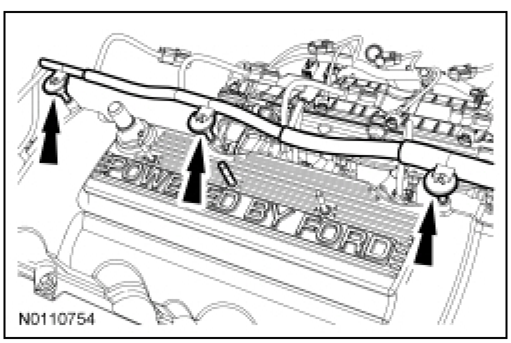
98. Connect the LH and RH CMP electrical connectors.



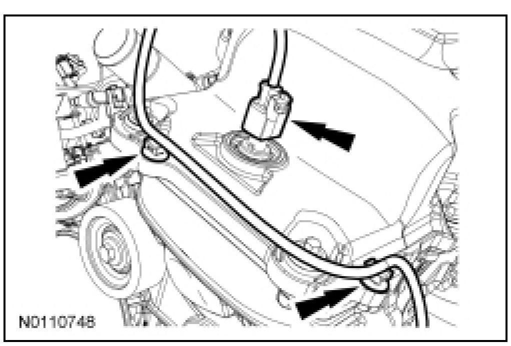
99. Connect the wiring harness retainer, the KS and Cylinder Head Temperature (CHT) sensor electrical connectors.



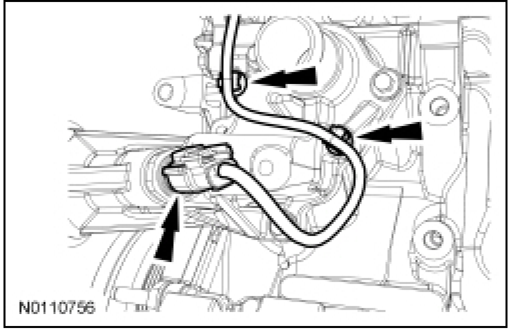
100. Connect the 3 engine wiring harness position retainers to the LH valve cover.



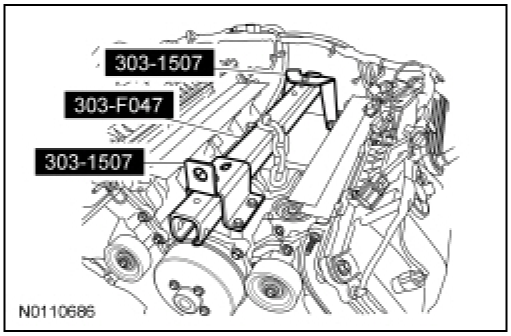
101. Connect the LH VCT variable force solenoid electrical connector and the 2 wiring harness retainers.



102. Connect the Engine Oil Pressure (EOP) switch electrical connector and wiring harness retainers.



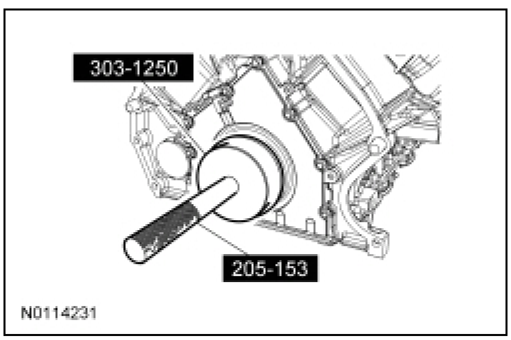
103. Install the Engine Lifting Bracket and Engine Support Brackets.



104. Using a suitable floor crane, remove the engine from the engine stand.

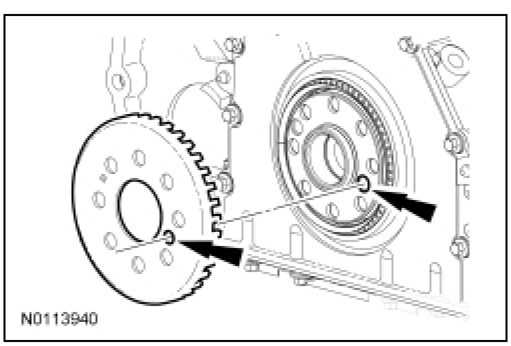
105. **NOTE:** *Lubricate the crankshaft rear seal with clean engine oil prior to installation.*

Using the Rear Main Seal Installer and Handle, install a new crankshaft rear seal.



106. **NOTE:** *Inspect the ignition pulse ring for damage. If the ignition pulse ring has been dropped or has any visual damage, it must be discarded.*

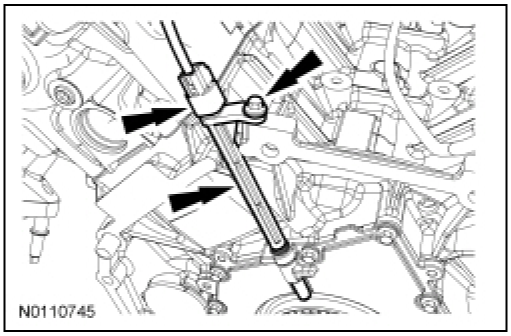
With the inset hole on the crankshaft ignition pulse ring aligned with the inset hole on the crankshaft flange, install the crankshaft ignition pulse ring.



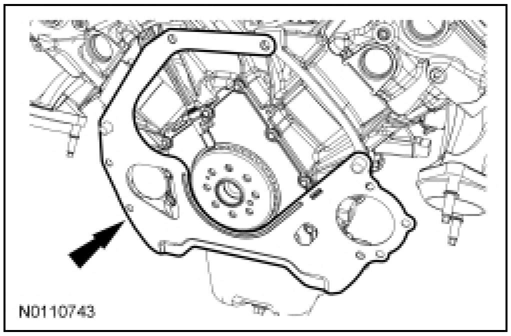
107. **NOTICE: The Crankshaft Position (CKP) sensor must be positioned into the fitting on the crankshaft rear seal retainer plate and be flush against the boss on the engine block before the bolt is installed. If the CKP sensor is installed incorrectly, the CKP sensor can be damaged.**

Position the Crankshaft Position (CKP) sensor, install the bolt and connect the electrical connector.

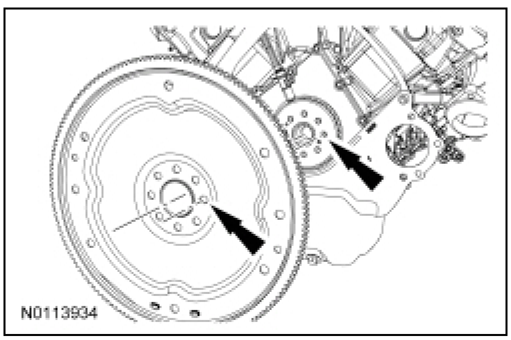
- Tighten to 10 Nm (89 lb-in).



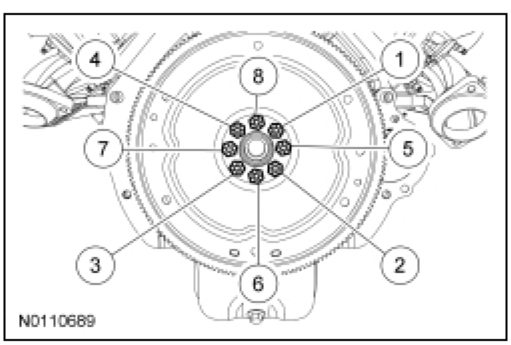
108. Remove the engine rear cover plate.



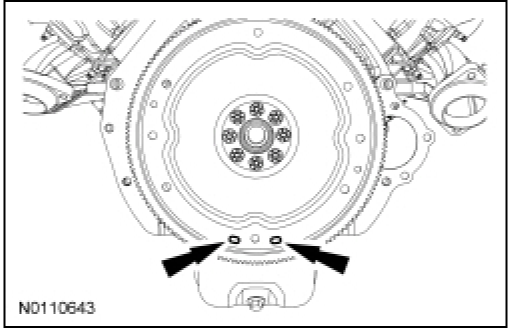
109. With the offset hole on the crankshaft ignition pulse ring aligned with the offset hole on the flexplate, position the flexplate and install 8 new bolts finger-tight.



110. Tighten the 8 new bolts in 6 stages.
    - Stage 1: Tighten fasteners 1 through 4 in sequence to 20 Nm (177 lb-in).
    - Stage 2: Tighten fasteners 1 through 4 in sequence to 35 Nm (26 lb-ft).
    - Stage 3: Tighten fasteners 1 through 4 in sequence an additional 60 degrees.
    - Stage 4: Tighten fasteners 5 through 8 in sequence to 20 Nm (177 lb-in).
    - Stage 5: Tighten fasteners 5 through 8 in sequence to 35 Nm (26 lb-ft).
    - Stage 6: Tighten fasteners 5 through 8 in sequence an additional 60 degrees.



111. Rotate the crankshaft clockwise until the 2 flexplate slotted holes are at the 6 o'clock position as shown prior to engine installation.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